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ASYMMETRIC HALF-BRIDGE FLYBACK POWER CONVERTER AND METHOD OF OPERATING THE SAME

Abstract

An asymmetric half-bridge flyback power converter includes a transformer, a resonant circuit, a first switch, a second switch, a discharging switch, an energy storage capacitor, and a controller. The first switch and the second switch are connected in series at a node, and the resonant circuit is coupled between the node and the transformer. The discharging switch is coupled between the node and the energy storage capacitor. When the power converter operates in a discontinuous resonant mode, and before the first switch is turned on by the controller, the discharging switch is first turned on so that the energy storage capacitor stores energy to rise a current flowing to the node.

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Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to a power converter and a method of operating the same, and more particularly to an asymmetric half-bridge flyback power converter and a method of operating the same.

Description of Related Art

[0002] The statements in this section merely provide background information related to the present disclosure and do not necessarily constitute prior art.

[0003] In recent years, as consumer electronics products have gradually increased their requirements for adapter power and size, switching power converters have the characteristics of high efficiency and small size, which have replaced linear voltage regulators and become the mainstream of consumer electronics products. An asymmetric half-bridge flyback power converter has the advantage of low input current ripple, and its built-in current ripple elimination mechanism is used to further effectively reduce the input current ripple. Therefore, the asymmetric half-bridge flyback power converter has become a more suitable circuit structure for switching power converters.

[0004] Traditional asymmetric half-bridge flyback power converters usually use metal oxide semiconductors (MOSFETs) as switching components for a pulse width modulation (PWM). The reason is that the MOSFET has the characteristics of fast switching speed and low loss. However, if the parasitic capacitance of the MOSFET is not fully discharged when the switch is turned on, there will be a cross voltage across a drain to a source of the MOSFET. This cross voltage will cause a loss with the current flowing through the source of the MOSFET, which is called the switching loss of the MOSFET. Since in the technical field of power converters, the requirements for high efficiency in their operation are getting higher and higher, it is necessary to reduce the switching losses of MOSFETs as much as possible.

[0005] Therefore, how to design an asymmetric half-bridge flyback power converter and a method of operating the same to achieve zero-voltage switching at any time point in the power conversion process to reduce MOSFET switching losses has become a critical topic in this field.

SUMMARY

[0006] In order to solve the above-mentioned problems, the present disclosure provides an asymmetric half-bridge flyback power converter. The asymmetric half-bridge flyback power converter includes a transformer, a resonant circuit, a switching circuit, a zero-voltage switching circuit, and a controller. The transformer includes a primary-side winding, a secondary-side winding, and a magnetizing inductance. The primary-side winding is coupled to the secondary-side winding, and the magnetizing inductance is connected to the primary-side winding in parallel. The resonant circuit is coupled to the primary-side winding. The switching circuit includes a first switch and a second switch connected in series at a node, and the node is coupled to the resonant circuit. The zero-voltage switching circuit is coupled to the node, and includes a discharging switch, a diode, and an energy storage capacitor connected in series. The controller is coupled to the zero-voltage switching circuit, and the controller controls the first switch and the second switch to be turned on or turned off to generate an output voltage. When an output energy of the power converter is less than an energy threshold and before the first switch is turned on by the controller, the controller first turns on the discharging switch so that the energy storage capacitor stores energy to rise a current flowing to the node. In one embodiment, when the output energy is the output voltage and the output voltage is less than a voltage threshold, the controller first turns on the discharging switch before the first switch is turned on by the controller. When the output energy is

an output current of the power converter and the output current is less than a first current threshold, the controller first turns on the discharging switch before the first switch is turned on by the controller.

[0007] In one embodiment, the asymmetric half-bridge flyback power converter further includes an auxiliary winding and a voltage regulation circuit. The auxiliary winding is coupled to the primary-side winding. The voltage regulation circuit is coupled to the auxiliary winding. The auxiliary winding is coupled to the primary-side winding to generate a winding voltage, and the voltage regulation circuit supplies power to the controller according to the winding voltage.

[0008] In one embodiment, the energy storage capacitor is connected to the voltage regulation circuit in parallel, and the energy storage capacitor stores a working voltage provided by the voltage regulation circuit, and the controller receives the working voltage to operate.

[0009] In one embodiment, the first switch comprises a first parasitic capacitance, and a capacitance value of the first parasitic capacitance is less than an energy storage capacitance value of the energy storage capacitor.

[0010] In one embodiment, when the first switch is turned on and the second switch is turned off, the magnetizing inductance stores energy of a magnetizing current. When a current value of the magnetizing current is greater than or equal to a current threshold, the controller turns off the first switch so that a second parasitic capacitance of the second switch discharges toward the node.

[0011] In one embodiment, when the first switch is turned off and the second switch is turned on, the energy of the primary-side winding is transferred to the secondary-side winding and the auxiliary winding. When the first switch and the second switch are turned off and the magnetizing current releases energy to zero, the magnetizing inductance, the first parasitic capacitance, and the second parasitic capacitance generate a resonance with a resonant period.

[0012] In one embodiment, the resonant period is related to a magnitude of an output current of the power converter.

[0013] In one embodiment, the resonant circuit includes a resonant capacitor. When the discharging switch is turned on, the resonant capacitor charges the second parasitic capacitance, the energy storage capacitor, and the magnetizing inductance.

[0014] In one embodiment, when the current rises to a second current threshold, the controller turns off the discharging switch, and the second current threshold is related to a time when the first parasitic capacitance is completely discharged.

[0015] In order to solve the above-mentioned problems, the present disclosure provides a method of operating an asymmetric half-bridge flyback power converter. The power converter includes a transformer, a resonant circuit, a first switch, a second switch, a discharging switch, and an energy storage capacitor. The first switch and the second switch are connected in series at a node, and the resonant circuit is coupled between the node and the transformer. The discharging switch is coupled between the node and the energy storage capacitor. The method includes steps of: controlling the first switch and the second switch to be turned on or turned off according to a feedback signal to generate an output voltage; determining whether an output energy of the power converter is less than an energy threshold; first turning on the discharging switch when the output energy is less than the energy threshold and before the first switch is controlled to be turned on.

[0016] In one embodiment, when the output energy is the output voltage, the method further includes steps of: determining whether the output voltage is less than a voltage threshold; first turning on the discharging switch when the output voltage is less than the voltage threshold and before the first switch is controlled to be turned on.

[0017] In one embodiment, when the output energy is an output current of the power converter, the method further includes steps of: determining whether the output current is less than a first current threshold; first turning on the discharging switch when the output current is less than the first current threshold and before the first switch is controlled to be turned on.

[0018] In one embodiment, the method further includes steps of: storing energy of a magnetizing

current of the transformer when the first switch is turned on and the second switch is turned off; turning off the first switch when a current value of the magnetizing current is greater than or equal to a current threshold; discharging a second parasitic capacitance of the second switch toward the node.

[0019] In one embodiment, the method further includes steps of: transferring the energy of the transformer to a secondary-side winding and an auxiliary winding of the power converter; generating a resonance with a resonant period by the magnetizing inductance, a first parasitic capacitance of the first switch, and the second parasitic capacitance of the second switch when the first switch and the second switch are turned off and the magnetizing current releases energy to zero. The resonant period is related to a magnitude of an output current of the power converter.

[0020] In one embodiment, the method further includes steps of: charging the second parasitic capacitance, the energy storage capacitor, and the magnetizing inductance by a resonant capacitor of the resonant circuit; turning off the discharging switch when the current rises to a second current threshold. The second current threshold is related to a time when the first parasitic capacitance is completely discharged.

[0021] Therefore, the main purpose and effect of the present disclosure is to enable zero-voltage switching in the discontinuous resonant mode. The method is to connect a discharging switch and a diode in series at the node between the first switch and the second switch. Before the first switch is turned on, the discharging switch is turned on, and the resonant circuit is used to reversely charge the magnetizing inductance to increase the reverse current flowing into the node between the first switch and the second switch. Therefore, this reverse current can be used to discharge the parasitic capacitance of the first switch to achieve the effect of zero-voltage switching.

[0022] It is to be understood that both the foregoing general description and the following detailed description are exemplary, and are intended to provide further explanation of the present disclosure as claimed. Other advantages and features of the present disclosure will be apparent from the following description, drawings, and claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0023] The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawing as follows:

[0024] FIG. 1A is a block circuit diagram of an asymmetric half-bridge flyback power converter according to the present disclosure.

[0025] FIG. 1B is a timing waveform diagram of operating the asymmetric half-bridge flyback power converter in a continuous resonant mode according to the present disclosure.

[0026] FIG. 2 is a timing waveform diagram of operating the asymmetric half-bridge flyback power converter in a discontinuous resonant mode according to the present disclosure.

[0027] FIG. 3A to FIG. 3G are respectively current path diagrams of operating the asymmetric half-bridge flyback power converter in the discontinuous resonant mode according to the present disclosure.

[0028] FIG. 4 is a flowchart of a method of operating the asymmetric half-bridge flyback power converter according to the present disclosure.

DETAILED DESCRIPTION

[0029] Reference will now be made to the drawing figures to describe the present disclosure in detail. It will be understood that the drawing figures and exemplified embodiments of present disclosure are not limited to the details thereof.

[0030] Please refer to FIG. 1A, which shows a block circuit diagram of an asymmetric half-bridge flyback power converter according to the present disclosure. The asymmetric half-bridge flyback

power converter **100** (hereinafter abbreviated as “power converter **100**”) receives a direct-current (DC) voltage V_{dc} , and converts the DC voltage V_{dc} into an output voltage V_o to supply power to a load **200** coupled to a rear end. The power converter **100** includes a power conversion circuit **1**, an auxiliary power supplying circuit **2**, a controller **3**, and a zero-voltage switching circuit **4**. The power conversion circuit **1** includes a switch circuit **12**, a resonant circuit **14**, a transformer **16**, and a rectification circuit **18**. In particular, there are various implementations of the rectification circuit **18**, and here the rectification circuit **18** includes an output capacitor C_o and an output diode D_o as an example. The switch circuit **12** receives the DC voltage V_{dc} , and the switch circuit **12** includes a first switch **S1** and a second switch **S2** connected in series. The transformer **16** includes a primary-side winding **162**, a secondary-side winding **164**, and a magnetizing inductance L_m . The primary-side winding **162** is coupled to the secondary-side winding **164**, and the magnetizing inductance L_m is connected to the primary-side winding **162** in parallel. In particular, the magnetizing inductance L_m may be an equivalent inductance formed by the primary-side winding **162**, or an additional winding wound by the transformer **16**.

[0031] One terminal of the resonant circuit **14** is coupled to a node SW between the first switch **S1** and the second switch **S2**, and the other terminal of the resonant circuit **14** is coupled to the primary-side winding **162**. In one embodiment, the resonant circuit **14** may include a resonant inductor L_r and a resonant capacitor C_r . One terminal of the resonant inductor L_r is coupled to the node SW between the first switch **S1** and the second switch **S2**, and the other terminal of the resonant inductor L_r is coupled to one terminal of the primary-side winding **162**. One terminal of the resonant capacitor C_r is coupled to the second switch **S2** and a ground terminal, and the other terminal of the resonant capacitor C_r is coupled to the other terminal of the primary-side winding **162**. In one embodiment, it is not limited to the structure of the resonant circuit **14** mentioned above, that is, any circuit structure that can enable the resonant circuit **14** to produce a resonant function should be included in the scope of this embodiment. One terminal of the rectification circuit **18** is coupled to the secondary-side winding **164**, and the other terminal of the rectification circuit **18** is coupled to the load **200** to provide the output voltage V_o to supply power to the load **200**.

[0032] The auxiliary power supplying circuit **2** includes an auxiliary winding **22** and a voltage regulation circuit **24**, and the auxiliary winding **22** is coupled to the primary-side winding **162**. One terminal of the voltage regulation circuit **24** is coupled to the auxiliary winding **22**, and the other terminal of the voltage regulation circuit **24** is coupled to the controller **3**. The auxiliary winding **22** generates a winding voltage V_{aux} by coupling the primary-side winding **162**, and the voltage regulation circuit **24** supplies power to the controller **3** according to a working voltage V_{CC} provided by the winding voltage V_{aux} . The zero-voltage switching circuit **4** is coupled to the node SW between the first switch **S1** and the second switch **S2**, and the zero-voltage switching circuit **4** includes a discharging switch **42**, a diode **44**, and an energy storage capacitor **46** connected in series. The controller **3** is coupled to the first switch **S1** and the second switch **S2**, and provides a pulse-width modulation (PWM) signal to control turning on or turning off the first switch **S1** and the second switch **S2** so as to control the power conversion circuit **1** to convert the DC voltage V_{dc} into the output voltage V_o . Moreover, the controller **3** receives a feedback signal corresponding to the output voltage V_o , and adjusts the PWM signal according to the feedback signal so as to stabilize/regulate a voltage value of the output voltage V_o by adjusting the PWM signal. The output capacitor C_o stores the output voltage V_o to provide the output voltage V_o to supply power to the load **200**. In one embodiment, the voltage regulation circuit **24** may be, for example, but not limited to, a circuit that can stabilize/regulate the winding voltage V_{aux} , such as a low drop-out linear regulator (LDO) or a clamping circuit. For example, the simplest implementation of the voltage regulation circuit **24** can be a diode, used to reverse bias and cut off a negative voltage of the winding voltage V_{aux} .

[0033] On the other hand, the controller **3** is coupled to the discharging switch **42**. When an output

energy of the power converter **100** is less than an energy threshold, and before the first switch **S1** is turned on by the controller **3**, the controller **3** first provides a control signal S_c to turn on the discharging switch **42** so that the energy storage capacitor **46** stores energy to rise a current flowing to the node SW. In particular, the energy storage capacitor **46** may be a capacitor with a larger capacitance value, and an electrolytic capacitor is a preferred implementation. Specifically, the energy storage capacitor **46** is mainly used to rise the current flowing to the node SW through energy storage, and therefore its capacitance value needs to be larger to provide better energy storage means. In one embodiment, the controller **3** may be, for example but not limited to, a control chip such as a microcontroller (MCU) or a digital signal processor (DSP). However, the controller **3** is not excluded and may also be implemented by a control circuit, a logic circuit, and other devices.

[0034] On the other hand, since the energy storage capacitor **46** is preferably an electrolytic capacitor, a capacitor with a larger capacitance value in the power converter **100** can be used as the energy storage capacitor **46**. For example, but not limited to, the input capacitor, the output capacitor, the capacitor of the power supply terminal of the controller **3**, etc., and this embodiment uses the energy storage capacitor **46** coupled to the power supply terminal of the controller **3** as an illustrative example to save additional capacitor configuration costs and increase the power density of the power converter **100**. However, it does not rule out that the energy storage capacitor **46** is configured independently. In one embodiment, since the energy storage capacitor **46** is coupled to the power supply terminal of the controller **3**, the energy storage capacitor **46** is connected in parallel to the voltage regulation circuit **24** to store the working voltage VCC provided by the voltage regulation circuit **24**, and the controller **3** receives the working voltage VCC to operate.

[0035] Please refer to FIG. 1B, which shows a timing waveform diagram of operating the asymmetric half-bridge flyback power converter in a continuous resonant mode according to the present disclosure, and also refer to FIG. 1A. When the power converter **100** operates in a continuous resonant mode and the first switch **S1** is turned on, the DC voltage V_{dc} charges the magnetizing inductance L_m of the transformer **16** and the resonant capacitor C_r of the resonant circuit **14**. Therefore, a magnetizing current I_{lm} flowing through the magnetizing inductance L_m and a resonant current I_{cr} flowing through the resonant capacitor C_r gradually increase. When the second switch **S2** is turned on, energy is transferred from the resonant capacitor C_r on the primary side of the transformer **16** to the secondary side of the transformer **16**, and the waveform of a secondary-side current I_{sec} is like the half-sine wave current waveform of the resonant converter. Due to changes in the output wide voltage and output current I_o (herein collectively referred to as output energy), the turned-on time of the first switch **S1** will be affected. Since under this operation, the magnetizing inductance L_m will be clamped at nV_o (that is, n times the output voltage V_o) to perform flux leakage, and when the output energy changes, the flux leakage time will change.

[0036] In general, since the switching component is not an ideal component, the first switch **S1** includes a first parasitic capacitance C_{s1} , and the second switch **S2** includes a second parasitic capacitance C_{s2} . When the power converter **100** operates in the continuous resonant mode, the power converter **100** usually continues to turn on the second switch **S2** to reversely charge the magnetizing inductance L_m after the magnetizing inductance L_m is discharged to 0 A so that the excitation current I_{lm} reverse (that is, the direction of the current is the direction flowing to the node SW) so that the magnetizing current I_{lm} is reversed (that is, the current flows toward the node SW). Afterward, this reverse current is used to discharge the first parasitic capacitance C_{s1} of the first switch **S1** to achieve the effect of zero voltage switching. This continuous resonant mode, although used at full load and high voltage, can achieve zero voltage switching.

[0037] However, when the output current I_o or the output voltage V_o (i.e., the output energy) decreases, the current peak value of the magnetizing current I_{lm} of the magnetizing inductance L_m also decreases, thereby causing the operating frequency of the power converter **100** to be too high.

Also, the circulating energy of the resonant circuit **14** will also limit the overall efficiency of the power converter **100**. Therefore, in the case of low output energy, the asymmetric half-bridge flyback power converter **100** usually operates in the discontinuous resonant mode of the magnetizing current I_{lm} (that is, the controller **3** adjusts the operation mode to the discontinuous resonant mode). However, if the second switch **S2** in the discontinuous resonant mode is turned off when the magnetizing inductance L_m discharges to 0 A, there will not be enough reverse magnetizing current I_{lm} in the discontinuous resonant mode, and zero voltage switching may not be achieved, thereby resulting in additional switching losses.

[0038] Therefore, the main purpose and effect of the present disclosure is to enable zero-voltage switching in the discontinuous resonant mode. The method is to connect a discharging switch **42** and a diode **44** in series at the node SW between the first switch **S1** and the second switch **S2**. Before the first switch **S1** is turned on, the discharging switch **42** is turned on, and the resonant circuit **14** is used to reversely charge the magnetizing inductance to increase the reverse current flowing into the node SW between the first switch **S1** and the second switch **S2**. Therefore, this reverse current can be used to discharge the parasitic capacitance of the first switch **S1** to achieve the effect of zero-voltage switching.

[0039] Furthermore, since under the condition of low output energy, the asymmetric half-bridge flyback power converter **100** will operate in the discontinuous resonant mode of the magnetizing current I_{lm} , the resonant circuit **14** must be used to reversely charge the magnetizing inductance to increase the reverse current flowing into the node SW between the first switch **S1** and the second switch **S2**. Therefore, the controller **3** must detect the level of output energy. In particular, the output energy may be the output voltage V_o . When the output energy is the output voltage V_o , and the output voltage V_o is less than the voltage threshold (for example, but not limited to, the output voltage V_o with a wide-range upper limit of 48V, the voltage threshold may be set at 12V). Before the first switch **S1** is turned on by the controller, the controller **3** first turns on the discharging switch **42** to use the resonant circuit **14** to reversely charge the magnetizing inductance to increase the current flowing into the node SW in the reverse direction. On the other hand, the output energy may be the output current I_o . When the output energy is the output current I_o and the output current I_o is less than a first current threshold (for example, but not limited to, for a fully loaded output current I_o of 20A, and the first current threshold may be set to 5 A), before the first switch **S1** is turned on by the controller **3**, the controller first turns on the discharging switch **42** to use the resonant circuit **14** to reversely charge the magnetizing inductance to increase the current flowing into the node SW in the reverse direction.

[0040] Please refer to FIG. 2, which shows a timing waveform diagram of operating the asymmetric half-bridge flyback power converter in a discontinuous resonant mode according to the present disclosure; please refer to FIG. 3A to FIG. 3G, which show respectively current path diagrams of operating the asymmetric half-bridge flyback power converter in the discontinuous resonant mode according to the present disclosure, and also refer to FIG. 1A to FIG. 1B. During time t_1 to time t_2 (refer to FIG. 3A), the controller **3** controls the first switch **S1** to be turned on, and controls the second switch **S2** and the discharging switch **42** to be turned off so that the output diode **Do** and the diode **44** are reverse biased and cut off. In this condition, the DC voltage V_{dc} charges the magnetizing inductance L_m , and the magnetizing inductance L_m stores the magnetizing current I_{lm} . Until the current value of the magnetizing current I_{lm} rises to greater than or equal to the current threshold, the controller **3** controls the first switch **S1** to be turned off to enter the time t_2 to time t_3 (i.e., dead time). During time t_2 to time t_3 (refer to FIG. 3B), the controller **3** controls the first switch **S1**, the second switch **S2**, and the discharging switch **42** to be turned off, and the output diode **Do** and the diode **44** are reverse biased and cut off. In this condition, the magnetizing current I_{lm} causes the second parasitic capacitance C_{s2} of the second switch **S2** to discharge toward the node SW, and after a junction diode of the second switch **S2** is forward biased and turned on, the dead time is ended and the controller **3** controls the second

switch **S2** to be turned on to enter the time t_3 to time t_4 .

[0041] During time t_3 to time t_4 (refer to FIG. 3C), the controller **3** controls the second switch **S2** to be turned on, and controls the first switch **S1** and the discharge switch **42** to be turned off. In this condition, the output diode **Do** is forward biased and turned on, and the diode **44** is reverse biased and cut off. Furthermore, the energy of the primary-side winding **162** is transferred to the secondary-side winding **164** and the auxiliary winding **22** to provide the remaining energy to the output capacitor **Co** and the controller **3** to maintain the normal operation of the power converter **100**. During time t_4 to time t_5 (refer to FIG. 3D), the controller **3** controls the first switch **S1**, the second switch **S2**, and the discharging switch **42** to be turned off so that the output diode **Do** and the diode **44** are reverse biased and cut off and the magnetizing current I_{lm} is released to **0A**.

[0042] During time t_5 to time t_6 (refer to FIG. 3E), the controller **3** continuously controls the first switch **S1**, the second switch **S2**, and the discharging switch **42** to be turned off, and the output diode **Do** and the diode **44** are continuously reverse biased and cut off. In this condition, the magnetizing inductance L_m , the first parasitic capacitance C_{s1} , and the second parasitic capacitance C_{s2} generate resonance with a resonant period (that is, the current flows back and forth through the magnetizing inductance L_m , the first parasitic capacitance C_{s1} , and the second parasitic capacitance C_{s2}), and the resonant period is related to a magnitude of the output current I_o of the power converter **100**. In general, when the output current I_o is larger, the resonant period is shorter, and vice versa. After a sufficient number of cycles of resonance occurs and the resonant current I_{cr} resonates to **0 A**, the controller **3** controls the discharge switch **42** to be turned on to enter the time t_6 to time t_7 .

[0043] During time t_6 and time t_7 (refer to FIG. 3F), the controller **3** controls the discharging switch **42** to be turned on so that the diode **44** is forward biased and turned on. Similarly, the controller **3** continuously controls the first switch **S1** and the second switch **S2** to continuously be turned off, and the output diode **Do** is continuously reverse biased and cut off. In this condition, the resonant capacitor **Cr** simultaneously charges the second parasitic capacitance C_{s2} of the second switch **S2**, the energy storage capacitor **46**, and the magnetizing inductance L_m . Since the capacitance value of the energy storage capacitor **46** will be much larger than the second parasitic capacitance C_{s2} of the second switch **S2**, most of the charging current will flow to the energy storage capacitor **46** so that the second parasitic capacitance C_{s2} of the second switch **S2** will not be fully charged immediately. Therefore, during the charging process, the magnetizing inductance L_m is charged with a large enough reverse current, and the direction of this current is opposite to the direction of the magnetizing current I_{lm} and flows through the node **SW** to the second parasitic capacitance C_{s2} to charge the second parasitic capacitor **Cs2**.

[0044] When a sufficiently large reverse current I_{sw} is reached (that is, when the current I_{sw} flowing in the direction of the node **SW** increases to a second current threshold- I_{th}), the controller controls the discharging switch **42** to be turned off to enter the time t_7 and time t_8 . During time t_7 and time t_8 (refer to FIG. 3G), the controller **3** controls the first switch **S1**, the second switch **S2**, and the discharging switch **42** to be turned off, and the output diode **Do** and the diode **44** are reverse biased and cut off. In this condition, the reverse current I_{sw} that is opposite to the magnetizing current I_{lm} discharges the first parasitic capacitance C_{s1} of the first switch **S1** so that a junction diode of the first switch **S1** is forward biased and turned on so as to achieve the effect of zero-voltage switching.

[0045] In particular, the above “large enough” means a current that is enough to completely charge the second parasitic capacitance C_{s2} of the second switch **S2** and completely discharge the first parasitic capacitance C_{s1} of the first switch **S1**. Therefore, the capacitance value of the first parasitic capacitance C_{s1} and the capacitance value of the second parasitic capacitance C_{s2} may preferably be smaller than the energy storage capacitance of the energy storage capacitor **46**, and the second current threshold- I_{th} is related to the time when the discharge of the first parasitic capacitance C_{s1} is completed. When the capacitance value of the first parasitic capacitance C_{s1}

and the capacitance value of the second parasitic capacitance Cs2 are larger, the second current threshold-Ith may be increased (referring to the higher its absolute value), and vice versa, and its formula is as follows:

[00001] $\sqrt{\frac{(Cs1 + Cs2) \times Vdc^2}{Lm}}$ [0046] In particular, -I_{lm} is the current setting value opposite to the magnetizing current I_{lm} (i.e., the second current threshold-I_{th}), and its setting value may be set by the controller 3.

[0047] Please refer to FIG. 4, which shows a flowchart of a method of operating the asymmetric half-bridge flyback power converter according to the present disclosure, and also refer to FIG. 1 to FIG. 3G. The power converter 100 includes a transformer 16, a resonant circuit 14, a first switch S1, a second switch S2, a discharging switch 42, and an energy storage capacitor 46. The first switch S1 and the second switch S2 are connected in series at a node SW, and the resonant circuit 14 is coupled between the node SW and the transformer 16. The discharging switch 42 is coupled between the node SW and the energy storage capacitor 46. The first switch S1, the second switch S2, and the discharging switch 42 are controlled by the controller 3. The method of operating the power converter 100 includes steps of: controlling the first switch and the second switch to be turned on or turned off according to a feedback signal to generate an output voltage (S100). In one embodiment, a pulse-width modulation (PWM) signal is provided by the controller 3 to control the first switch S1 and the second switch S2 to be turned on or turned off so as to control the power conversion circuit 1 to convert a direct-current (DC) voltage V_{dc} into an output voltage V_o. Moreover, the controller 3 receives the feedback signal corresponding to the output voltage V_o, and adjusts the PWM signal according to the feedback signal so as to stabilize/regulate a voltage value of the output voltage V_o by adjusting the PWM signal.

[0048] Afterward, determining whether an output energy of the power converter is less than an energy threshold (S120). Due to changes in the output wide voltage and output current I_o (herein collectively referred to as output energy), when the output energy changes, the flux leakage time will change. Therefore, it is necessary to determine whether the power converter 100 is operating in the continuous resonant mode or the discontinuous resonant mode by whether the output energy is lower than the threshold so as to provide corresponding operation steps according to the different modes. When the determination result in step (S120) is yes, it means that the output energy is lower than an energy threshold and the power converter 100 operates in the discontinuous resonant mode. Therefore, first turning on the discharging switch when the output energy is less than the energy threshold and before the first switch is controlled to be turned on (S140). The resonant circuit 14 can be used to reversely charge the magnetizing inductance L_m to increase the reverse current flowing into the node SW between the first switch S1 and the second switch S2, and use this reverse current to discharge the parasitic capacitance of the first switch S1 to achieve the effect of zero-voltage switching.

[0049] When the determination result in step (S120) is no, it means that the output energy is greater than the energy threshold and the power converter 100 operates in the continuous resonant mode. Therefore, return to step (S100) for continued determination. In one embodiment, the operating steps not detailed in FIG. 4 may be referred to FIG. 1 to FIG. 3G and will not be repeated here. In addition, in an embodiment of the present disclosure, the control of the first switch S1, the second switch S2, and the discharging switch 42 is not limited to being controlled only by the controller 3. All controllers, control circuits (composed of passive electronic components), logic circuits (composed of logic components) and other devices that can perform the above operations should be included in the scope of this embodiment.

[0050] Although the present disclosure has been described with reference to the preferred embodiment thereof, it will be understood that the present disclosure is not limited to the details thereof. Various substitutions and modifications have been suggested in the foregoing description, and others will occur to those of ordinary skill in the art. Therefore, all such substitutions and

modifications are intended to be embraced within the scope of the present disclosure as defined in the appended claims.

Claims

1. An asymmetric half-bridge flyback power converter comprising: a transformer comprising a primary-side winding, a secondary-side winding, and a magnetizing inductance; the primary-side winding coupled to the secondary-side winding, and the magnetizing inductance connected to the primary-side winding in parallel, a resonant circuit coupled to the primary-side winding, a switching circuit comprising a first switch and a second switch connected in series at a node, and the node coupled to the resonant circuit, a zero-voltage switching circuit coupled to the node, and comprising a discharging switch, a diode, and an energy storage capacitor connected in series, and a controller coupled to the zero-voltage switching circuit, and the controller configured to control the first switch and the second switch to be turned on or turned off to generate an output voltage, wherein when an output energy of the power converter is less than an energy threshold and before the first switch is turned on by the controller, the controller is configured to first turn on the discharging switch so that the energy storage capacitor stores energy to rise a current flowing to the node.
2. The asymmetric half-bridge flyback power converter as claimed in claim 1, wherein when the output energy is the output voltage and the output voltage is less than an a voltage threshold, the controller is configured to first turn on the discharging switch before the first switch is turned on by the controller; when the output energy is an output current of the power converter and the output current is less than a first current threshold, the controller is configured to first turn on the discharging switch before the first switch is turned on by the controller.
3. The asymmetric half-bridge flyback power converter as claimed in claim 1, further comprising: an auxiliary winding coupled to the primary-side winding, and a voltage regulation circuit coupled to the auxiliary winding, wherein the auxiliary winding is coupled to the primary-side winding to generate a winding voltage, and the voltage regulation circuit is configured to supply power to the controller according to the winding voltage.
4. The asymmetric half-bridge flyback power converter as claimed in claim 3, wherein the energy storage capacitor is connected to the voltage regulation circuit in parallel, and the energy storage capacitor is configured to store a working voltage provided by the voltage regulation circuit, and the controller is configured to receive the working voltage to operate.
5. The asymmetric half-bridge flyback power converter as claimed in claim 4, wherein the first switch comprises a first parasitic capacitance, and a capacitance value of the first parasitic capacitance is less than an energy storage capacitance value of the energy storage capacitor.
6. The asymmetric half-bridge flyback power converter as claimed in claim 5, wherein when the first switch is turned on and the second switch is turned off, the magnetizing inductance stores energy of a magnetizing current, wherein when a current value of the magnetizing current is greater than or equal to a current threshold, the controller is configured to turn off the first switch so that a second parasitic capacitance of the second switch discharges toward the node.
7. The asymmetric half-bridge flyback power converter as claimed in claim 6, wherein when the first switch is turned off and the second switch is turned on, the energy of the primary-side winding is transferred to the secondary-side winding and the auxiliary winding, wherein when the first switch and the second switch are turned off and the magnetizing current releases energy to zero, the magnetizing inductance, the first parasitic capacitance, and the second parasitic capacitance generate a resonance with a resonant period.
8. The asymmetric half-bridge flyback power converter as claimed in claim 7, wherein the resonant period is related to a magnitude of an output current of the power converter.
9. The asymmetric half-bridge flyback power converter as claimed in claim 7, wherein the resonant

circuit comprises a resonant capacitor; when the discharging switch is turned on, the resonant capacitor charges the second parasitic capacitance, the energy storage capacitor, and the magnetizing inductance.

10. The asymmetric half-bridge flyback power converter as claimed in claim 9, wherein when the current rises to a second current threshold, the controller is configured to turn off the discharging switch, and the second current threshold is related to a time when the first parasitic capacitance is completely discharged.

11. A method of operating an asymmetric half-bridge flyback power converter, the power converter comprising a transformer, a resonant circuit, a first switch, a second switch, a discharging switch, and an energy storage capacitor; the first switch and the second switch connected in series at a node, and the resonant circuit coupled between the node and the transformer; the discharging switch coupled between the node and the energy storage capacitor; the method comprising steps of: controlling the first switch and the second switch to be turned on or turned off according to a feedback signal to generate an output voltage, determining whether an output energy of the power converter is less than an energy threshold, and first turning on the discharging switch when the output energy is less than the energy threshold and before the first switch is controlled to be turned on.

12. The method of operating the asymmetric half-bridge flyback power converter as claimed in claim 11, wherein when the output energy is the output voltage, further comprising steps of: determining whether the output voltage is less than a voltage threshold, and first turning on the discharging switch when the output voltage is less than the voltage threshold and before the first switch is controlled to be turned on.

13. The method of operating the asymmetric half-bridge flyback power converter as claimed in claim 11, wherein when the output energy is an output current of the power converter, further comprising steps of: determining whether the output current is less than a first current threshold, and first turning on the discharging switch when the output current is less than the first current threshold and before the first switch is controlled to be turned on.

14. The method of operating the asymmetric half-bridge flyback power converter as claimed in claim 11, further comprising steps of: storing energy of a magnetizing current of the transformer when the first switch is turned on and the second switch is turned off, turning off the first switch when a current value of the magnetizing current is greater than or equal to a current threshold, and discharging a second parasitic capacitance of the second switch toward the node.

15. The method of operating the asymmetric half-bridge flyback power converter as claimed in claim 14, further comprising steps of: transferring the energy of the transformer to a secondary-side winding and an auxiliary winding of the power converter, and generating a resonance with a resonant period by the magnetizing inductance, a first parasitic capacitance of the first switch, and the second parasitic capacitance of the second switch when the first switch and the second switch are turned off and the magnetizing current releases energy to zero, wherein the resonant period is related to a magnitude of an output current of the power converter.

16. The method of operating the asymmetric half-bridge flyback power converter as claimed in claim 15, further comprising steps of: charging the second parasitic capacitance, the energy storage capacitor, and the magnetizing inductance by a resonant capacitor of the resonant circuit, and turning off the discharging switch when the current rises to a second current threshold, wherein the second current threshold is related to a time when the first parasitic capacitance is completely discharged.
